

Git for Scientists

So that we can still **prove what is human** when
working with AI in research.

Lars Schöbitz, Global Health Engineering, ETH Zurich, and SwissRN local node ETH Zurich. Slides: ghe-open.ch/slides 



2023

<https://ghe-open.ch/slides/>

Digital Research in 2023

- Git was hard
- Manuscripts live in DOCX
- Track changes was the only record of who did what, and it vanishes on accept



Poll

Has AI written part of a document you produced this month?

👋 Hands up for yes 👋

Poll

Who can prove which parts of the document were generated by AI, and which parts by a human?

👋 Hands up for yes 👋

How do you prove it?

Ask yourself:

1. From memory?
2. Self documented?
3. Automated?
4. From the traces you leave?



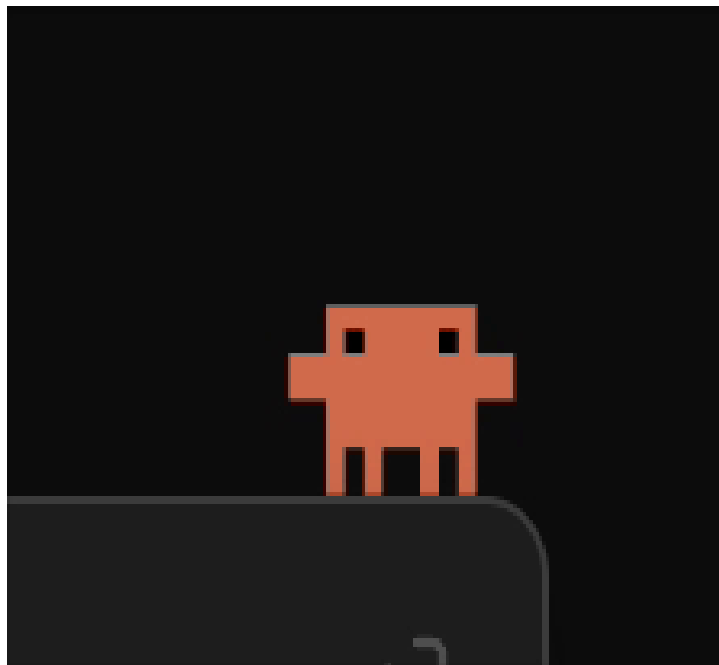
2026

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(Digital) Research in 2026

AI makes Git easier to use, and more necessary.

- Git is a language prompt away
- Tell the AI to track changes
- Plan well with GitHub issues
- Empower yourself with agentic AI
- Stay in control



Declaration of AI

International Science Council: A global reporting standard.

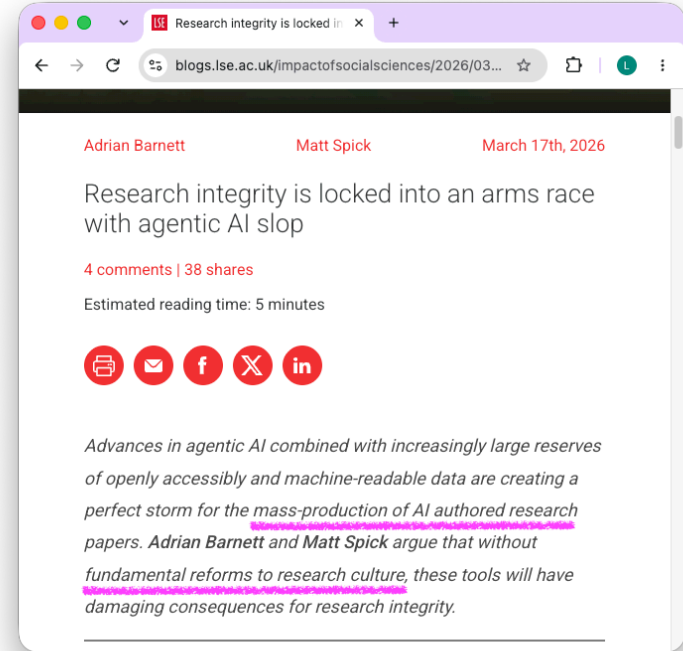
Proposed taxonomy

Based on the results of Round 1, a review of existing AI disclosure standards, and discussion within the core team, we propose 18 controlled categories (a “taxonomy”) to serve as a framework for disclosing AI use in research tasks. The proposed list of research tasks is:

1. *Idea generation*
2. *Hypothesis development*
3. *Literature search*
4. *Literature summarization*
5. *Design > Study design / experiment design*
6. *Design > Data collection methodology*
7. *Design > Data analysis plan*
8. *Data collection (operations / activities)*
9. *Data creation*
10. *Computer code generation/refinement/auditing*
11. *Data cleaning/preprocessing/auditing*
12. *Quantitative data analysis*
13. *Qualitative data analysis*
14. *Data visualization*
15. *Figure or image generation*
16. *Drafting manuscript or abstract text*
17. *Translation*
18. *Editing or rewriting including of reference lists*

Declaration of human

- A paper in 30 minutes ([Spick et al. 2026](#))
- Well documented open data makes it possible
- How do we prove our work? ([Barnett and Spick 2026](#))



Git for Scientists: teaching the need to know

So that we can prove what is human when working with AI in research.

- Read the tracked changes between two versions of a file (a diff)
- Understand how to experiment safely (a branch)
- Review what has gone into a set of changes (a pull request)
- Know what should never enter a repository, or an AI tool (data security)

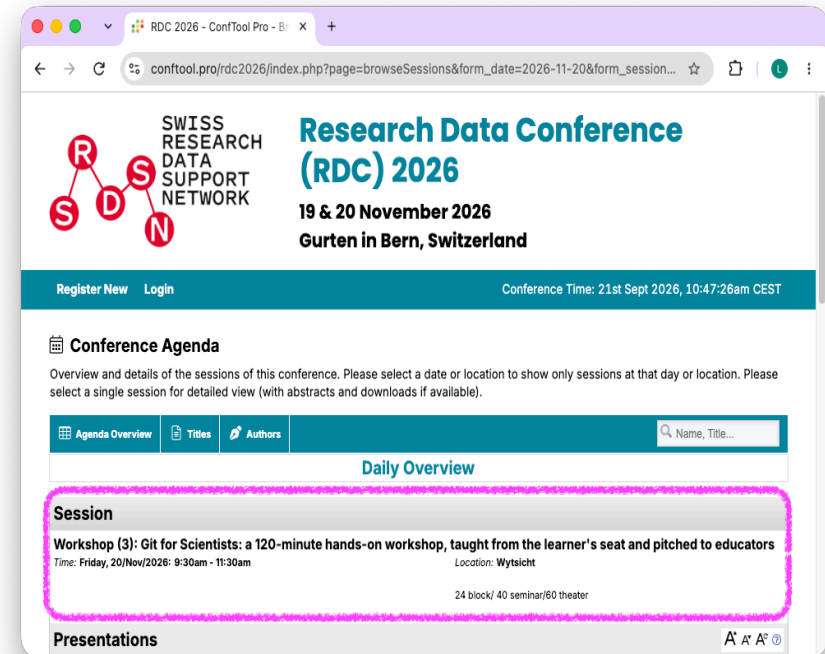
Git for Scientists: open educational resource

- Handbook for how to set up and run the workshop
- Open, CC BY, for any data steward or RSE to teach
- gitforsci.github.io/website 



Git for Scientists: RDC 2026, Bern

- Research Data Conference, 19 and 20 November 2026, Gurten, Bern
- 120-minute hands-on workshop, taught from the learner's seat and pitched to educators
- Friday 20 November, 9:30 to 11:30
- [Abstract](#)  and [ConfTool session](#) 



Agents for Scientists

<https://ghe-open.ch/slides/>[↑]

Agents for Scientists

Handing the need to know over to the AI

- **Work safely:** Git lets you revert
- **Experiment freely:** agentic AI opens endless possibilities
- **Track AI use:** prompts and resulting changes are recorded

First full-day workshop on 29 September.

FAIR by doing

- 15 participants, 12 months, 5 workshops
- Git, GitHub, FAIR data publishing, agentic AI
- One day with Greg Wilson: [Organizational Change for Open Science](#) 
- Anyone is welcome, especially qualitative researchers and people from non-IT backgrounds

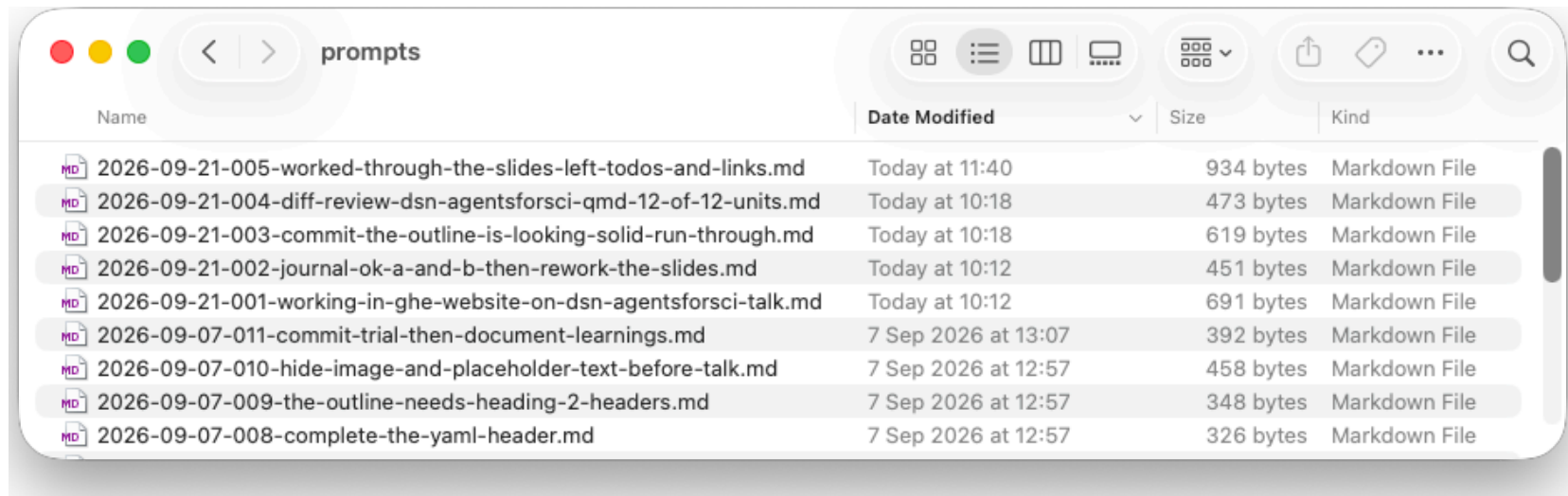
Express interest here:



Declaration of AI use

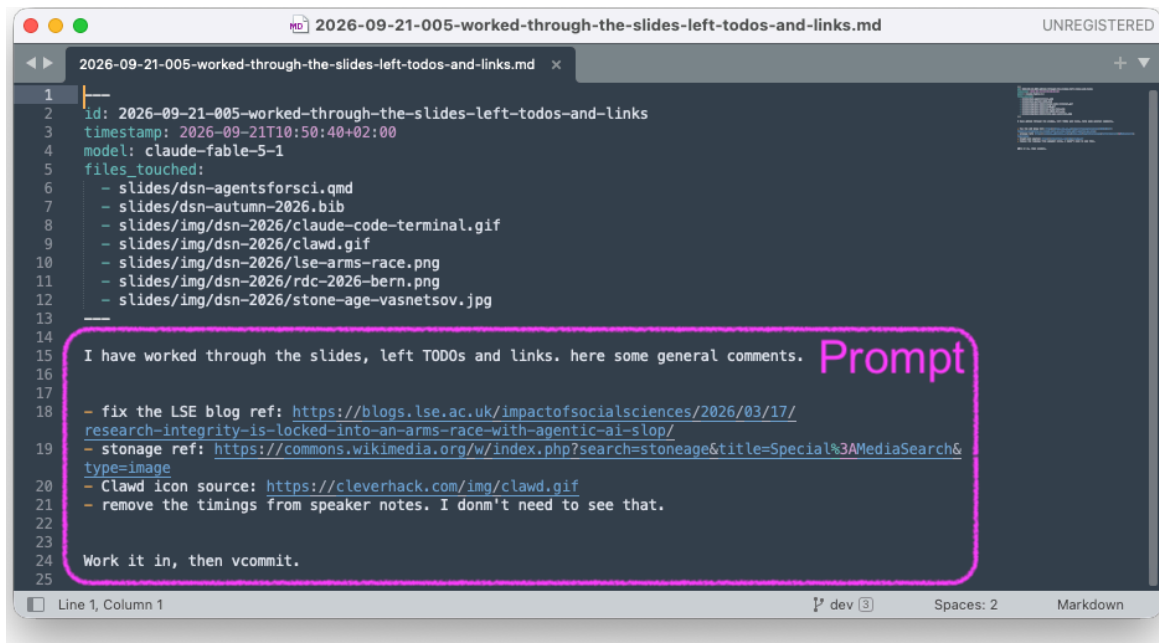
<https://ghe-open.ch/slides/>

A prompts folder in every repository



Every prompt recorded, one Markdown file each. This is the prompts folder of these slides.

One file per prompt

A screenshot of a code editor window titled '2026-09-21-005-worked-through-the-slides-left-todos-and-links.md'. The editor shows a markdown file with a list of files touched and a prompt. The prompt is highlighted with a pink dashed border and the word 'Prompt' in pink. The prompt text is: 'I have worked through the slides, left TODOs and links. here some general comments. Prompt' followed by a list of tasks: '- fix the LSE blog ref: https://blogs.lse.ac.uk/impactofsocialsciences/2026/03/17/research-integrity-is-locked-into-an-arms-race-with-agentic-ai-slop/', '- stonage ref: https://commons.wikimedia.org/w/index.php?search=stoneage&title=Special%3AMediaSearch&type=image', '- Clawd icon source: https://cleverhack.com/img/clawd.gif', and '- remove the timings from speaker notes. I donm't need to see that.' The prompt ends with 'Work it in, then vcommit.'

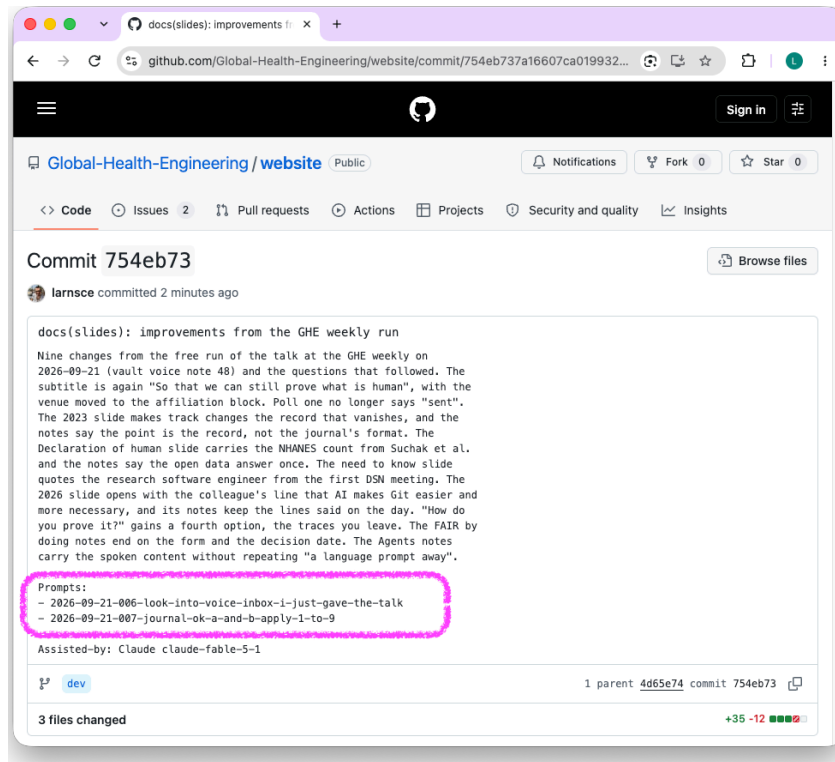
```
1 ---
2 id: 2026-09-21-005-worked-through-the-slides-left-todos-and-links
3 timestamp: 2026-09-21T10:50:40+02:00
4 model: claude-fable-5-1
5 files_touched:
6   - slides/dsn-agentsforsci.qmd
7   - slides/dsn-autumn-2026.bib
8   - slides/img/dsn-2026/claude-code-terminal.gif
9   - slides/img/dsn-2026/clawd.gif
10  - slides/img/dsn-2026/lse-arms-race.png
11  - slides/img/dsn-2026/rdc-2026-bern.png
12  - slides/img/dsn-2026/stone-age-vasnetsov.jpg
13 ---
14
15 I have worked through the slides, left TODOs and links. here some general comments. Prompt
16
17 - fix the LSE blog ref: https://blogs.lse.ac.uk/impactofsocialsciences/2026/03/17/
18   research-integrity-is-locked-into-an-arms-race-with-agentic-ai-slop/
19 - stonage ref: https://commons.wikimedia.org/w/index.php?search=stoneage&title=Special%3AMediaSearch&
20   type=image
21 - Clawd icon source: https://cleverhack.com/img/clawd.gif
22 - remove the timings from speaker notes. I donm't need to see that.
23
24 Work it in, then vcommit.
25
```

An id and timestamp. The used AI model. The files that were touched.

<https://ghe-open.ch/slides/>

A commit skill

- Claude Code makes the commit
- Standardised
- The prompt files behind it are named
- Contribute at: [Global-Health-Engineering/skills](https://ghe-open.ch/slides/Global-Health-Engineering/skills) 

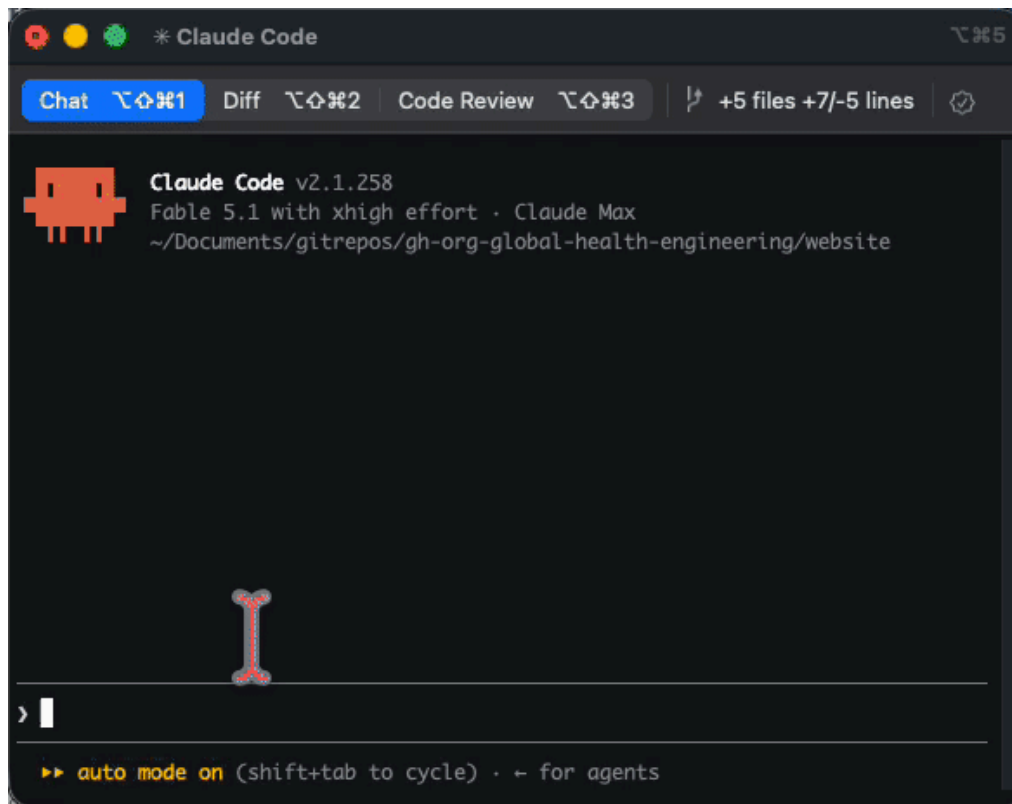


Let the AI declare your AI use

Prompt:

“For my latest DSN slides, use the traces of commits, prompt folder, pull requests, etc. to declare the use of AI against the [ETH Zurich Information Security and Artificial Intelligence \(AI\) Guide for Users of AI at ETH Zurich \(24 July 2025\)](#)  and the [ETH Library "Generative AI in the citation rules \(17 February 2025\)"](#) . Use only what's present in this repository.”

10 minutes: Get up and walk



The declaration, from the record

- **Tool:** Claude Code, model claude-fable-5-1, on 21 September 2026
- **Record:** 10 prompt files, 11 commits, 6 with an Assisted-by trailer
- **AI:** built the deck from outline and recording, notes, images, commits
- **Human:** outline, the talk twice, 26 review decisions, 5 own commits
- **Data:** public repository, CC BY 4.0, no personal or confidential data
- **Not recorded:** recordings, tool settings, ETH list of approved services

Full declaration: ghe-open.ch/slides/dsn-agentsforsci-ai-declaration 

Thanks

REFERENCES

- **These slides:** [download as PDF](#) 
- **AI use:** [declaration from the record](#) 
- Built with [Quarto](#)  and reveal.js
- **Section backgrounds:** image taken from [Danielle Navarro](#) 

Funding and support: [SwissRN](#)  local node ETH Zurich and the [Data Stewardship Network](#)  at ETH Zurich, funded by the [swissuniversities Open Science Programme](#) , and [Global Health Engineering](#) , ETH Zurich

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RESOURCES

- [Git for Scientists handbook](#) 
- [Agents for Scientists](#) 
- [FAIR by doing proposal](#) 
- [FAIR by doing: expression of interest form](#) 

References

- Barnett, Adrian, and Matt Spick. 2026. *Research Integrity Is Locked into an Arms Race with Agentic AI Slop*. LSE Impact of Social Sciences blog.
<https://blogs.lse.ac.uk/impactofsocialsciences/2026/03/17/research-integrity-is-locked-into-an-arms-race-with-agentic-ai-slop/> ↗.
- Spick, Matt, Tulsi Suchak, and Adrian Barnett. 2026. *The 30 Minute Manuscript: An Example Scientific Paper by AI Agent*. Octopus, preprint.
<https://doi.org/10.57874/cm47-c165> ↗.